

Series of exercises 1: The real numbers

Exercise 1 Let E and F two subsets of $\mathbb{R}$ such that $E \cap F \neq \emptyset$. Show that

1. If E is a bounded set then, we have $\inf E \leq \sup E$.
2. If E and F are bounded sets, then $E \cap F$ is a bounded set.
3. $\sup(E \cap F) \leq \min(\sup E, \sup F)$ and $\max(\inf E, \inf F) \leq \inf(E \cap F)$.

Exercise 2 Let E and F two subsets of $\mathbb{R}$ such that $E = -F$. Show that

1. The set E is bounded above if the set F is bounded below.
2. $\sup E = -\inf F$.

Exercise 3 Let A and B two subsets of $\mathbb{R}$. We define the subset $A + B = \{a + b; a \in A, b \in B\}$. Show that

1. If A and B are bounded sets then $A + B$ is also a bounded set.
2. $\inf(A + B) = \inf A + \inf B$.
3. $\sup(A + B) = \sup A + \sup B$.

Exercise 4 Compute $\sup$, $\inf$, $\max$ and $\min$ of the given subsets of $\mathbb{R}$.

$$A = \{1, 2, 3, 4\}, \quad B =]1, 4], \quad C = [0, 1[\quad \text{et} \quad D = [0, 1].$$

Exercise 5 Find the supremum $\sup E$ and infimum $\inf E$ of each set E . State whether they are in E

1. $E = \left\{ x \in \mathbb{R} : x = -\frac{1}{n} + (1 + (-1)^n), n \geq 1 \right\}$.
2. $E = \{x \in \mathbb{R} : x^2 < 9\}$.

3. $E = \{x \in \mathbb{Z} : x^2 < 9\}$. 4. $E = \{x \in \mathbb{R} : |2x + 1| \leq 5\}$.
5. $E = \left\{x \in \mathbb{Q} : \frac{1}{2} < (x^2 + 1)^{-1}\right\}$. 6. $E = \left\{\frac{1}{n} + \sin\left(\frac{2n\pi}{3}\right) : n \in \mathbb{N}^*\right\}$.

Exercise 6 Let A , B and C three subsets of the set of the real numbers $\mathbb{R}$, such that

$$A = \left\{\frac{n-1}{n}, n \in \mathbb{N}^*\right\}, \quad B = \left\{\frac{2n+1}{n+1}, n \in \mathbb{N}\right\} \text{ et } C = A \cup B.$$

1. Show that A , B and C are bounded sets.
2. Compute the supremum $\sup C$ and the infimum $\inf C$ if they exist.

Exercise 7 Let a nondecreasing function $f : [0, 1] \longrightarrow [0, 1]$; that is for all $x, y \in [0, 1]$, $x < y \implies f(x) \leq f(y)$. Let $A = \{x \in [0, 1], x \leq f(x)\}$

1. Show that A has an upper bound a . Deduce that $a \leq f(a)$.
2. Show that $a \geq f(a)$. Deduce that the function f admits at least one fixed point.

Exercise 8 Show that for all real numbers x and y , the absolute value $|\cdot|$ satisfies the following properties

1. $|x| = \max\{x, -x\}$.
2. $|x| \leq M$ is equivalent to $-M \leq x \leq M$, where M is a positive real number.
3. $||x| - |y|| \leq |x + y| \leq |x| + |y|$ and $|xy| = |x| |y|$.
4. $\max\{x, y\} = \frac{1}{2}(x + y + |x - y|)$ and $\min\{x, y\} = \frac{1}{2}(x + y - |x - y|)$.

Exercise 9 Write the following expressions in equivalent forms not involving absolute values.

1. $a + b + |a - b|$.

2. $|a + b - 2c + |a - b||$.
3. $a + b + 2c + |a - b| + |a + b - 2c + |a - b||$.

Exercise 10

1. Show that for all real numbers x and y , the integer part of a real number $[.]$ satisfies the following properties
 - (a) $[x] + [y] \leq [x + y] \leq [x] + [y] + 1$.
 - (b) $\forall n \in \mathbb{N} : [x + n] = [x] + n$.
 - (c) $[x] \leq x < [x + 1] = [x] + 1$.
 - (d) $[x] - 1 < x < [x] + 1$.
 - (e) $0 \leq [2x] - 2[x] \leq 1$.
2. Compute $[x]$ and write x depending on $[x]$ for the real numbers $x = 0.5$, $x = 3$, $x = -\frac{13}{5}$ and $x = -6.15$.
3. Draw the graph of $f(x) = [x]$, for all real numbers x such that $|x| \leq 5$.

Exercise 11 The real $x = \sqrt{\sqrt{2} + \sqrt{3}}$ is a rational number or an irrational number?

Exercise 12 Prove the following insertions by the mathematical induction

1. For all $n \in \mathbb{N}$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.
2. For all $n \in \mathbb{N}$, $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(n+1)(4n^2-1)}{3}$.
3. If $x_1, x_2, \dots, x_n$ are arbitrary real numbers, then

$$|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n|.$$